

1. [C]

program to create child pid and parent pid

```
#include <stdio.h>
#include <unistd.h>
int main() {
    pid_t pid = fork()
    if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
    } else {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
    }
    return 0;
}
```

OUTPUT :

child PID:8745

parent PID:8756

2. [b]

purpose of two file management system calls :

1. open() System Call

Purpose: Opens an existing file or creates a new file (if the appropriate flag is used).

It returns a file descriptor, which is used by other system calls such as read() and write().

2. write() System Call

Purpose: Writes data from the program to a file.

It takes the file descriptor, the data to be written, and the number of bytes as arguments.

3.[a]

The system calls required to copy the contents of one file to another are:

open() - Opens the source file for reading and the destination file for writing .

read() - Reads data from the source file into a buffer.

write() - Writes the data from the buffer to the destination file.

close() - Closes both the source and destination files after the copying process is complete.

pseudo code

Start

Open source file in read mode

Open destination file in write mode

While data is available

Read data from source file

```
Write data to destination file
Close source file
Close destination file
Stop
```

4.[a]

purpose of fork and exec systems calls :

fork() - Creates a child process from the parent process.

Example:

When we open a new Terminal window in Linux, the shell uses fork() to create a new process.

A web server uses fork() to create a separate process to handle each client request.

exec() - Replaces the current process with a new program and starts its execution.

Example :

After the shell creates a child process using fork(), it uses exec() to run a command such as ls, pwd, or gcc.

5.[a]

File management is an essential function of an operating system that ensures files are stored securely, accessed efficiently, and managed systematically.

File reading is the process of accessing data stored in a file using operating system system calls.

The open(), read(), and close() system calls are used to open a file, read its contents, and close it

```
#include <fcntl.h>
#include <unistd.h>
```

```
int main() {
    int fd;
    char ch;
```

```
    fd = open("student.txt", O_RDONLY);
```

```
    while (read(fd, &ch, 1) > 0)
        write(1, &ch, 1);
```

```
    close(fd);
```

```
    return 0;
}
```

EXAMPLE :

INPUT: RAJU
HARI

OUTPUT: RAJU
HARI